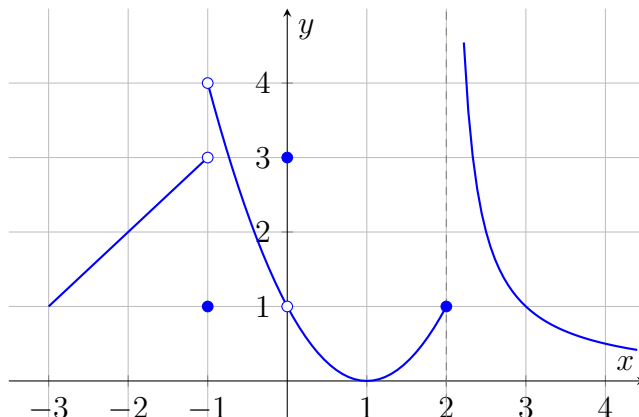


Math 1512 Calculus 1 Review
Fall 2026

1. Memorization:

- (a) The definition of “ f is continuous at $x = a$ ”
- (b) Both limit definitions of $f'(a)$, the derivative of f at $x = a$ (as $x \rightarrow a$ and as $h \rightarrow 0$)
- (c) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$ and $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta}$
- (d) Derivatives of $\sin(x)$, $\cos(x)$, $\tan(x)$, $\sec(x)$, $\csc(x)$, $\cot(x)$
- (e) Power rule, product rule, quotient rule, chain rule
- (f) The linearization $L(x)$ of f at $x = a$, and the differential dy
- (g) Definition of a critical number of f
- (h) State the Intermediate Value Theorem, the Extreme Value Theorem, Fermat's Theorem, and the Mean Value Theorem
- (i) First derivative test, second derivative test, and the definition of an inflection point
- (j) Antiderivatives of x^n ($n \neq -1$), $\sin(x)$, $\cos(x)$, $\sec^2(x)$, $\sec(x) \tan(x)$
- (k) The definite integral $\int_a^b f(x) dx$ as a limit of Riemann sums
- (l) State both parts of the Fundamental Theorem of Calculus
- (m) The average value of f on $[a, b]$
- (n) Volume of a cylinder, a cone, and a sphere. Area of a circle. Surface area of a sphere.

2. Limits from a graph. Let g be the function graphed below. A filled dot is a point on the graph, an open dot is not.



- $g(-1)$, $\lim_{x \rightarrow -1^-} g(x)$, $\lim_{x \rightarrow -1^+} g(x)$, $\lim_{x \rightarrow -1} g(x)$
- $g(0)$ and $\lim_{x \rightarrow 0} g(x)$
- $\lim_{x \rightarrow 2^-} g(x)$, $\lim_{x \rightarrow 2^+} g(x)$, $\lim_{x \rightarrow 2} g(x)$
- At which x values in $(-3, 4)$ is g not continuous? For each, say whether the discontinuity is removable, a jump, or infinite.
- At which of these x values is g continuous from the left? From the right?
- Suppose $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} h(x) = -2$. Use the limit laws to evaluate $\lim_{x \rightarrow a} [3f(x) - 5h(x)]$, $\lim_{x \rightarrow a} f(x)h(x)$, $\lim_{x \rightarrow a} \frac{f(x)}{h(x)}$, and $\lim_{x \rightarrow a} \left(\sqrt{f(x)} + h(x)^2 \right)$.

3. Computing limits. Evaluate the limit, or explain why it does not exist. Use ∞ or $-\infty$ when appropriate.

(a) $\lim_{x \rightarrow 2} \frac{x^2+x-6}{x^2-4}$

(b) $\lim_{x \rightarrow -3} \frac{x+3}{x^2-9}$

(c) $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$

(d) $\lim_{h \rightarrow 0} \frac{(2+h)^2-4}{h}$

(e) $\lim_{x \rightarrow 2} \frac{\frac{1}{x}-\frac{1}{2}}{x-2}$

(f) $\lim_{r \rightarrow 0} \frac{\sqrt{4+r}-\sqrt{4-r}}{r}$

(g) $\lim_{x \rightarrow 3^+} \frac{x+1}{3-x}$

(h) $\lim_{x \rightarrow 2^-} \frac{x+2}{x^2+x-6}$

(i) $\lim_{x \rightarrow -2} \frac{6}{x^2-4}$ (consider the one-sided limits)

(j) $\lim_{t \rightarrow 3} \frac{t^3}{(t-1)(t-3)^2}$

4. Squeeze Theorem.

(a) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{3}{x}\right)$

(b) $\lim_{x \rightarrow \infty} \frac{3+\cos(x)}{x^2}$

(c) If $2x-1 \leq f(x) \leq x^2-2x+3$ for all x , find $\lim_{x \rightarrow 2} f(x)$.

(d) Explain why we cannot use the product law to evaluate the limit in (a).

5. Limits at infinity and asymptotes. Evaluate the limit.

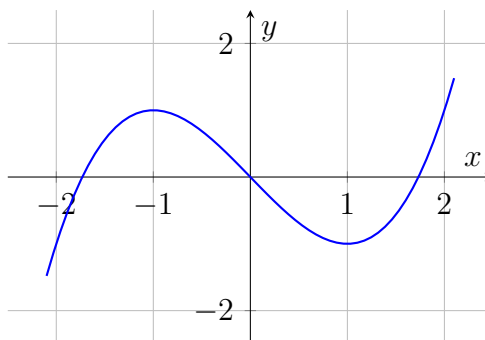
- (a) $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 2}{5x^2 + 4}$
- (b) $\lim_{x \rightarrow -\infty} \frac{2x^3 + 1}{x^4 - x}$
- (c) $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + x}}{2x - 1}$
- (d) $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 1}}{x + 3}$ (Recall $\sqrt{x^2} = |x|$)
- (e) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 6x} - x)$
- (f) Find all vertical and horizontal asymptotes of $f(x) = \frac{6x - 3x^2}{x^2 - 16}$.
- (g) Find all vertical and horizontal asymptotes of $f(x) = \frac{x^2 - x - 2}{x^2 - 4}$. Why is $x = 2$ not a vertical asymptote?

6. Continuity and the Intermediate Value Theorem.

- (a) Give an example of a function $f(x)$ and a value $x = a$ where f has a removable discontinuity. Sketch the graph.
- (b) Give an example of a function $f(x)$ and a value $x = a$ where f has a non-removable discontinuity and $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \infty$. Sketch the graph.
- (c) Give an example of a function $f(x)$ and a value $x = a$ where f has a non-removable discontinuity and $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$.
- (d) Let $f(x) = \begin{cases} x + 2 & x < -1 \\ x^2 & -1 \leq x \leq 1 \\ 3 - x & x > 1 \end{cases}$. Sketch the graph of f . Use the definition of continuity to decide whether f is continuous at $x = -1$ and at $x = 1$. At which x values is f not differentiable?
- (e) Where is $g(x) = \frac{\sqrt{x-3}}{x^2-25}$ continuous?
- (f) Find c so that $f(x) = \begin{cases} \frac{x^2}{x+2} & x \leq 2 \\ 3x + c & x > 2 \end{cases}$ is continuous at $x = 2$.
- (g) Find a and b so that $f(x) = \begin{cases} 2 & x < 1 \\ ax + b & 1 \leq x < 4 \\ 8 & x \geq 4 \end{cases}$ is continuous everywhere.
- (h) Is $f(x) = \begin{cases} \frac{x^2-9}{x-3} & x \neq 3 \\ 5 & x = 3 \end{cases}$ continuous at $x = 3$? Use the definition.
- (i) Show that $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.
- (j) Show that $\cos(x) = x$ has a solution in $(0, \frac{\pi}{2})$.

7. The derivative as a limit.

- (a) Find the equation of the line tangent to $f(x) = \sqrt{x}$ at $x = 4$.
- Sketch the graph of $f(x)$. Plot the point $(4, f(4))$, and draw the line tangent to this point. Draw another point $(x, f(x))$ and find the slope of the secant line between these two points. Set up a limit of the slopes of secant lines that represents the slope of the tangent line.
 - Evaluate the limit in part (i) to determine the slope of the tangent line.
 - Write an equation for the tangent line.
- (b) Use the limit definition of the derivative to find $f'(x)$ if $f(x) = 3x^2 - x + 4$.
- (c) Use the limit definition of the derivative to find $f'(x)$ if $f(x) = \frac{2}{x^2}$.
- (d) Use the limit definition of the derivative to find $f'(x)$ if $f(x) = \sqrt{2x+1}$. Then find the tangent line at $(4, 3)$.
- (e) Use the limit definition of the derivative to find $f'(x)$ if $f(x) = x^3 + 2$.
- (f) A ball's height is $h(t) = 200 - 16t^2$ feet after t seconds. Find the slope of the secant line between $t = 1$ and $t = 3$, with units. Then find the instantaneous velocity at $t = 2$ as a limit.
- (g) Is $f(x) = \sqrt{x}$ differentiable at $x = 0$? Why or why not? Reference what it means for a function to be differentiable. What is the line tangent to $f(x)$ at $x = 0$?
- (h) Each limit is the derivative of some function f at some number a . Find f and a , then evaluate. $\lim_{h \rightarrow 0} \frac{(1+h)^3 - 1}{h}$ and $\lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x - 3}$.
- (i) If $f(x) = \frac{1}{x}$, draw a graph of $f'(x)$ without calculating it explicitly.
- (j) Sketch the graph of f' for the function f graphed below. Where is f' zero, positive, negative?



8. Differentiation rules. You may use any of the derivative rules.

- (a) Find $f'(x)$ if $f(x) = 6x^4 - \frac{2}{x^2} + 3\sqrt[3]{x} - 7$
- (b) Find $g'(x)$ if $g(x) = (x^3 + 2x)(4 - x^2)$
- (c) Find y' if $y = \frac{x^2+1}{2x+1}$
- (d) Find $h'(x)$ if $h(x) = \frac{A}{x^2} + \frac{B}{x^4}$, where A and B are constants
- (e) Use quotient rule to find the derivative of $g(s) = \frac{2s}{s^2+1}$. Then use product rule to find it again, and check that your answers agree.
- (f) At what points (x, y) is the tangent line to $g(x) = \frac{x}{x^2+4}$ horizontal?
- (g) Find the tangent line to $f(x) = x^3 - 3x + 1$ at $x = 2$. Find all points where the tangent line is horizontal.
- (h) A particle has position $s(t) = t^3 - 9t^2 + 24t$ meters at time t seconds. Find the velocity $v(t)$ and acceleration $a(t)$, with units. When is the particle at rest?

9. Trig derivatives.

- (a) Using the derivatives of $\sin(x)$ and $\cos(x)$, calculate the derivatives of $\tan(x)$, $\sec(x)$, $\csc(x)$, and $\cot(x)$.
- (b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$, $\lim_{x \rightarrow 0} \frac{\tan(2x)}{x}$, and $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(5x)}$.
- (c) Find $k'(x)$ if $k(x) = x^2 \sin x + 3x + 2$
- (d) Find $f'(\theta)$ if $f(\theta) = \theta^2 \cos \theta - 4 \tan \theta$
- (e) Find y' if $y = \frac{\sec x}{1 + \tan x}$
- (f) Find the tangent line to $y = x \cos x$ at the point $(\pi, -\pi)$.
- (g) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = x + 2 \cos x$ has a horizontal tangent line.

10. Chain rule.

- (a) Given a composition of functions $f(x) = (g \circ h)(x) = g(h(x))$, what does the chain rule say about $f'(x)$?
- (b) Use the table to find $f'(2)$ for $f(x) = g(h(x))$. Then find $f'(2)$ for $f(x) = g(x)h(x)$.

x	-1	0	1	2	3	4
$h(x)$	2	1	0	3	-1	5
$h'(x)$	-1	2	1	4	0	-2
$g(x)$	0	-2	1	5	3	-1
$g'(x)$	3	0	-1	5	2	4

- (c) Find the equation of the line tangent to $x(t) = \sin(\sin(\sin(t)))$ when $t = \pi$
- (d) Find the equation of the line tangent to $h(t) = (1 + 2t^2)^4$ when $t = 0$
- (e) Find $f'(x)$ if $f(x) = \sqrt{x^4 + 3x^2 + 5}$
- (f) Find $g'(x)$ if $g(x) = 3x^2 \sec(x^3 - 1)$
- (g) Find $h'(x)$ if $h(x) = \frac{\cos^2(x)}{(2x+3)^3}$
- (h) Find y' if $y = \sin((3x^2 - 1)^4)$
- (i) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = 2 \sin x + \cos(2x)$ has a horizontal tangent line.

11. Implicit differentiation.

- (a) Sketch the curve $x^2 + y^2 = 4$. Use implicit differentiation to find $\frac{dy}{dx}$ as a function of x and y . Where are the horizontal and vertical tangent lines? Answer from the formula for $\frac{dy}{dx}$, not just the picture.
- (b) Find the equation of the tangent line at $(\pi/8, \pi/8)$ for the curve $\tan(x + y) + \sec(x - y) = 2$
- (c) Find the tangent line to $x^3 + y^3 = 9xy$ at $(2, 4)$.
- (d) Find the tangent line to $x^2y + y^3 = 2$ at $(1, 1)$.
- (e) For the curve $y^2 = x^3 - 3x + 2$, find $\frac{dy}{dx}$, the tangent line at $(2, 2)$, and all points with a horizontal tangent line.

12. Related rates. Draw a picture, name the quantities that change, and write the equation relating them before differentiating.
- (a) If x, y, z are implicitly functions of t , and $x(1 - y) + 5z^3 = y^2z^2 + x^2 - 3$, find z' when $x = 4, y = -2, z = 1, x' = 9, y' = -3$
 - (b) A 13 ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/s. How fast is the top sliding down when the bottom is 5 ft from the wall?
 - (c) A tank is an inverted cone with radius 3 m at the top and height 6 m. Water is pumped in at $4 \text{ m}^3/\text{min}$. How fast is the water level rising when the water is 4 m deep?
 - (d) The height of a cylinder increases at $3 \text{ cm}/\text{min}$ while its radius decreases at $0.5 \text{ cm}/\text{min}$. How fast is the volume $V = \pi r^2 h$ changing when $r = 2 \text{ cm}$ and $h = 4 \text{ cm}$? Is it increasing or decreasing?
 - (e) A rectangle's length increases at $3 \text{ in}/\text{min}$ and its width decreases at $2 \text{ in}/\text{min}$. When the length is 10 in and the width is 4 in, how fast is the area changing? Is it increasing or decreasing?
 - (f) A marble moves along the curve $x^2 + xy + y^2 = 9$ (x, y in feet). At $(3, 0)$, x is decreasing at 2 ft/s. Is y increasing or decreasing, and how fast?
13. Linear approximation and differentials.
- (a) Draw the graph of a function $f(x)$. Also draw $L(x) = f'(a)(x - a) + f(a)$. For x close to a , $f(x) \approx L(x)$. Use this to approximate $\Delta f = f(a + \Delta x) - f(a)$. Label Δf and your approximation for it in your picture.
 - (b) If $f(1) = 1$ and $f'(1) = \frac{1}{2}$, without calculating the linearization of f at 1, what do you expect $f(\frac{3}{2})$ to be? Now calculate the linearization and verify your answer.
 - (c) Find the linearization of $f(x) = \sqrt{x+5}$ at $a = 4$. Use it to approximate $\sqrt{9.1}$.
 - (d) Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$. Use it to approximate $\sqrt[3]{27.27}$.
 - (e) Find the linearization of $f(x) = \sin x$ at $a = \frac{\pi}{6}$. Use it to approximate $\sin(\frac{\pi}{6} + 0.02)$.
 - (f) The radius of a sphere is measured as 10 cm with a possible error of 0.05 cm. Use differentials to estimate the maximum error in the computed volume.

14. Extreme values. Use the closed interval method to find the absolute maximum and minimum values.

- (a) Summarize the closed interval method for finding the absolute extrema of a function over a closed interval.
- (b) Give an example of a function that has a critical number a with $f'(a) = 0$, but $(a, f(a))$ is not a local max or min.
- (c) Give an example of a function that has a local max or min even though it has no horizontal tangent lines.
- (d) $f(x) = 2x^3 - 9x^2 + 12x + 1$ on $[0, 3]$
- (e) $f(x) = x^4 - 8x^2 + 3$ on $[-1, 3]$
- (f) $f(x) = \frac{4x}{x^2+4}$ on $[-4, 0]$
- (g) $f(x) = \sqrt{9 - x^2}$ on $[-1, 3]$
- (h) $f(x) = 2 \cos x + \sin(2x)$ on $[0, \pi]$

15. Mean Value Theorem.

- (a) Find every c guaranteed by the MVT for $f(x) = x^2 - 4x + 1$ on $[1, 5]$.
- (b) Find every c guaranteed by the MVT for $f(x) = \sqrt{x}$ on $[0, 4]$.
- (c) If $f(2) = -1$ and $f'(x) \leq 3$ for all x , how large can $f(6)$ be?
- (d) Show that $f(x) = |x|$ on $[-1, 1]$ has no c with $f'(c) = \frac{f(1)-f(-1)}{1-(-1)}$. Which hypothesis of the MVT fails?
- (e) Show that $x^3 + 5x - 2 = 0$ has exactly one real solution.

16. Shape of a graph. Find the intervals of increase and decrease, the local extrema, the intervals of concavity, and the inflection points.

(a) $f(x) = x^3 - 6x^2 + 9x + 1$

(b) $f(x) = 3x^4 - 4x^3$

(c) $f(x) = x + 2 \sin x$ on $[0, 2\pi]$

(d) Use the second derivative test to classify the critical numbers of $f(x) = x^4 - 2x^2$.

(e) Sketch a function with $f' > 0$ and $f'' < 0$ everywhere. Sketch one with $f' < 0$ and $f'' > 0$ everywhere.

(f) Assume a is a positive constant, c is a negative constant, and b is a constant. Let $f(x) = x(ax^2 + bx + c)$, a cubic through $(0, 0)$ whose inflection point is the origin.

i. See what $f''(0) = 0$ means, then use this to simplify $f(x)$.

ii. Use the quadratic formula to find the roots of $f(x)$.

iii. Find the roots of $f'(x)$.

iv. Draw the graph of f . Label the local max and min by $(x_0, f(x_0))$ for each critical number x_0 you found.

17. Curve sketching. Find the domain, intercepts, asymptotes, intervals of increase and decrease, local extrema, concavity, and inflection points. Then sketch the graph and label all of these.

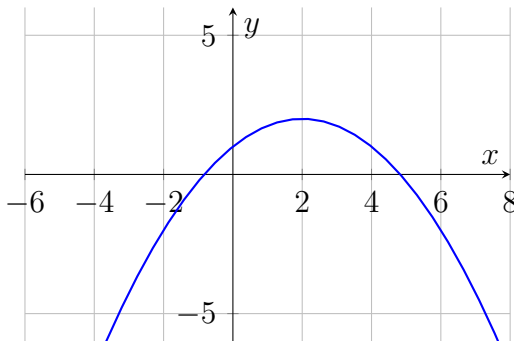
(a) $f(x) = \frac{1}{x^2+3}$

(b) $f(x) = \frac{x^2-1}{x^2-4}$

(c) $f(x) = (x+2)(x-1)^2$

(d) $f(x) = \frac{x+1}{x-2}$

18. Optimization. Draw a picture, introduce variables, write the quantity to optimize as a function of one variable, and verify that your answer is a max or min.
- A box is twice as long as it is wide. Find the dimensions that maximize the volume of the box, if you have 40 square feet of material to build it.
 - A farmer has 2000 feet of fencing to build a rectangular field, fenced on 3 sides, along a straight river. What dimensions maximize the area?
 - Maximize the product xy if the point (x, y) is on the upper half of the unit circle.
 - A box with a square base and an open top is made from 300 cm^2 of material. Find the largest possible volume.
 - A rectangular pen is split in half by a divider parallel to one side. The outer fence costs \$4 per meter and the divider costs \$2 per meter. With \$200, what is the largest possible area?
 - Find the points on the parabola $y = x^2$ closest to the point $(0, 2)$.
19. Antiderivatives and initial value problems. Find the most general antiderivative.
- $4x^3, \quad 2x^{-3}, \quad \sqrt{x^3}, \quad x^{-3/4}, \quad \sin x, \quad \cos x, \quad \sec^2(x), \quad \sec(x)\tan(x)$
 - $g(s) = \frac{4-2s+\sqrt{s^3}}{\sqrt{s}}$
 - $f(s) = \frac{s^3-s^2+s\sqrt{s}}{s^{3/2}}$
 - $h(x) = abcx + d^2ef^3x^2$, where a, b, c, d, e, f are constants
 - $f(x) = \sin(4x)$
 - If $f'(x) = x^2 - 1$ and $f(1) = 1$, find $f(x)$.
 - If $f''(t) = 2\cos t - 6\sin t$, $f(0) = 1$, and $f'(0) = 4$, find $f(t)$.
 - If $\frac{ds}{dt} = 4\sin(2t) - \cos(2t)$ and $s(\frac{\pi}{4}) = 3$, find $s(t)$.
 - A particle has acceleration $a(t) = 6t - 4$, initial velocity $v(0) = 3$, and initial position $s(0) = 1$. Find $s(t)$.
 - Carefully draw $y = 2x$ and 2 of its antiderivatives on the same set of axes. When a function has an x -intercept, what does this mean about the shape of its antiderivative?
 - Carefully draw $y = 1 - 3x^2$ and 2 different antiderivatives on the same set of axes. If a function has a local min or max, what does it mean about the shape of its antiderivative?
 - Graph an antiderivative of the function graphed below.

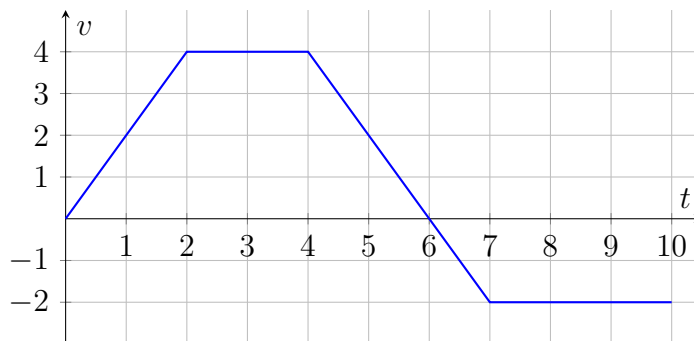


20. Riemann sums.

- Use 6 rectangles with the right endpoint rule to approximate the area under $\sin(x)$ over $[0, \pi]$. Graph $\sin(x)$ and draw each rectangle.
- Approximate the area under $f(x) = x^2$ on $[0, 4]$ with $n = 4$ rectangles using right endpoints, left endpoints, and midpoints. Which are over estimates, and which are under estimates?
- Use $n = 3$ rectangles with midpoints as sample points to approximate the area under $f(x) = x^2 + 1$ on $[0, 6]$.
- Write $\int_1^4 (x^2 + 3x) dx$ as a limit of Riemann sums using right endpoints. Do not evaluate.
- Write $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral, then evaluate it.

21. The definite integral as area. Evaluate by interpreting the integral in terms of areas. Do not use the Fundamental Theorem.

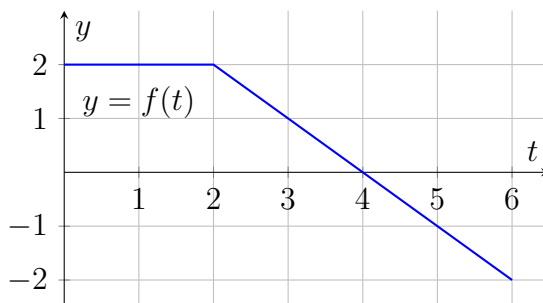
- $\int_0^2 (3x + \sqrt{4 - x^2}) dx$
- $\int_{-4}^4 (2 - \sqrt{16 - x^2}) dx$
- $\int_{-2}^2 (x^3 + \sqrt{4 - x^2}) dx$
- $\int_0^5 |2x - 4| dx$
- If $\int_0^5 f(x) dx = 7$ and $\int_0^2 f(x) dx = 3$, find $\int_2^5 f(x) dx$, $\int_5^2 f(x) dx$, and $\int_2^5 (3f(x) - 1) dx$.
- The velocity $v(t)$ (in m/s) of a particle moving on a line is graphed below, for t in seconds.



- What is the displacement of the particle from $t = 0$ to $t = 10$?
- What is the total distance traveled from $t = 0$ to $t = 10$?
- If the particle is at $x = 3$ m at $t = 0$, where is it at $t = 10$?
- What is the acceleration at $t = 5$? When is the particle farthest from where it started?

22. Fundamental Theorem of Calculus, Part 1. Find the derivative.

- (a) Calculate $f'(\pi/4)$ if $f(x) = \int_{-x}^x \cos(t)dt$, without calculating the antiderivative.
- (b) $g(x) = \int_1^x \sqrt{t^3 + 2} dt$
- (c) $g(x) = \int_{x^2}^4 t\sqrt{1+t^2} dt$
- (d) $F(x) = \int_1^{\sqrt{x}} \sin(s^2)ds$
- (e) $g(x) = \int_{2x}^{x^3} \frac{1}{1+t^4} dt$
- (f) Let $g(x) = \int_0^x f(t)dt$, where f is graphed below. Find $g(4)$, $g'(4)$, and $g''(4)$. Where on $[0, 6]$ does g have its maximum?



- (g) FTC part 1 says slopes undo areas: the rate the area $\int_0^x f(t)dt$ changes with an increase in x is equal to (*blank*). FTC part 2 says areas undo slopes. Fill in the blank, and explain why FTC part 2 makes sense.

23. Fundamental Theorem of Calculus, Part 2. Evaluate the integral.

- (a) Calculate $\int_{32}^{34} 2x dx$ as an area (without calculus). Then calculate it using part 2 of the Fundamental Theorem of Calculus.
- (b) $\int_1^4 \left(3\sqrt{x} - \frac{2}{x^2}\right) dx$
- (c) $\int_0^{\pi/4} \sec x \tan x dx$
- (d) $\int_0^{\pi} (\sin x + 2 \cos x) dx$
- (e) $\int_1^8 x^{-1/3} dx$
- (f) What is wrong with the calculation $\int_{-1}^2 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^2 = -\frac{3}{2}$?

24. U-sub. Clearly declare each substitution. For definite integrals, change the bounds to u -bounds.

- (a) $\int_1^2 x\sqrt{x-1}dx$
- (b) If $f'(t) = (\sin(t) + 1)^3 \cos(t)$, find the family of possible $f(t)$'s.
- (c) Demonstrate that u -substitution undoes chain rule by calculating $\int_a^b g'(h(x))h'(x)dx$.
- (d) $\int \frac{x^3}{(x^4+1)^2}dx$
- (e) $\int_0^1 x \cos\left(\frac{\pi x^2}{2}\right) dx$
- (f) $\int \sqrt{x} \sin(x^{3/2}) dx$
- (g) $\int \frac{\sin(1/x)}{x^2}dx$
- (h) $\int_0^4 \frac{x}{\sqrt{x^2+9}}dx$
- (i) $\int_{-2}^1 x\sqrt{x+3}dx$
- (j) $\int_{-\pi/2}^{\pi/2} x^2 \sin x dx$ (look for symmetry first)
- (k) Perform the substitution $u = 2x - 1$ in $\int_{1/2}^1 (2x - 1)dx$ without evaluating. Approximate the area represented by the x -integral and by the u -integral using 2 rectangles with right endpoints. Draw the picture for each.

25. Area between curves. Sketch the region and find its area.

- (a) $y = x$ and $y = x^2 - 6$
- (b) $y = x$ and $y = 5x - x^2$
- (c) $y = x^2$ and $y = 8 - x^2$. Set up the area integrating with respect to x , then with respect to y .
- (d) $x = y^2$ and $x = y + 2$ (integrate with respect to y)
- (e) $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$

26. Average value.

- (a) Find the average value of $f(x) = \frac{1}{x^2}$ on $[1, 3]$.
- (b) Find the average value of $f(x) = \sin x$ on $[0, \pi]$.
- (c) Find the average value of $f(x) = x^2$ on $[0, 3]$, and find c in $[0, 3]$ with $f(c)$ equal to the average value.

27. Volumes. Sketch the region and a representative slice. Set up an integral for the volume, then evaluate it.
- Set up (do not evaluate). Draw the relevant dV and where it lives in the total volume.
 - The region enclosed by the x -axis, $x = \pi$, and $y = \sin x$, revolved around the x -axis
 - The region enclosed by $y = 1$, $y = 4$, $y = x$, and $y = 2x + 1$, revolved around the y -axis
 - The region enclosed by $y = \sin x$ and $y = 1 - \sin x$, $x \in [0, \pi]$, revolved around $y = -1$
 - The region between $y = x$ and $y = x^2$, revolved around the x -axis
 - The region bounded by $y = x + 1$, $y = 0$, $x = 0$, and $x = 3$, revolved around the x -axis
 - The region bounded by $x = y^2$, $x = 0$, and $y = 2$, revolved around the y -axis
 - The region between $y = 3x$ and $y = x^2$, revolved around the y -axis, and then around the line $y = -1$
28. Work.
- If 2 J of work is needed to stretch a spring from 30 cm (natural length) to 42 cm, how much work is needed to stretch it from 35 cm to 40 cm? When the spring is pulled with 30 N of force, what will its length be?
 - A force of 12 N holds a spring stretched from its natural length of 10 cm to 14 cm. How much work is done stretching it from 12 cm to 16 cm?
 - A spring has natural length 10 cm and spring constant k . Find the work W_1 to stretch it from 10 cm to 15 cm and the work W_2 to stretch it from 15 cm to 20 cm, in terms of k . How are W_1 and W_2 related?
 - A force of $f(x) = 3\sqrt{x}$ pounds acts on an object x feet from the origin. Find the work done moving the object from $x = 1$ to $x = 4$.
 - The linear density of a wire is $\rho(x) = (x/100)^2 + 1$ g/cm for $0 \leq x \leq 100$, where x is the distance in cm from the light end. Find the total mass of the wire and its average density. The wire hangs over the edge of a building, heavy side down. How much work does it take to pull the wire to the roof? Use $F = mg$ with $g = 981$ cm/s².
 - A particle moves along a straight line, acted on by a force of $F(x) = (x - 1) \sin \left(2(x - 1)^2 - \frac{\pi}{4} \right)$ pounds when it is x feet from its starting position. How much work does the force do as the particle moves from its starting position to $x = 1$?